

PCS-302 : Data Structures Lab

Queue Using an Array | Create, EnQueue, DeQueue, IsEmpty, Size

Problem Statement

Design an algorithm and a program to implement a queue using an array. The program should implement the following queue operations: Create(), EnQueue(k), DeQueue(), IsEmpty() and Size().

Example

```
create()      --> Queue created, empty

enqueue(10)   --> Queue: 10
enqueue(20)   --> Queue: 10 20
enqueue(30)   --> Queue: 10 20 30

size()        --> 3

dequeue()     --> removes 10 (front), Queue: 20 30
```

Algorithm

```
Create():
    front = 0
    rear = -1
    count = 0

EnQueue(k):
    if rear == MAX - 1: print "Overflow", return
    rear = rear + 1
    arr[rear] = k
    count = count + 1

DeQueue():
    if count == 0: print "Underflow", return
    remove arr[front]
    front = front + 1
    count = count - 1

IsEmpty():
    return (count == 0)

Size():
    return count
```

C Program

```
#include <stdio.h>
#define MAX 5

int arr[MAX];
int front = 0;
int rear = -1;
```

```

int count = 0;

void create() {
    front = 0;
    rear = -1;
    count = 0;
    printf("Queue created.\n");
}

int isEmpty() {
    if (count == 0)
        return 1;
    else
        return 0;
}

void enqueue(int k) {
    if (rear == MAX - 1) {
        printf("Queue Overflow. Cannot enqueue %d.\n", k);
        return;
    }
    rear = rear + 1;
    arr[rear] = k;
    count = count + 1;
    printf("%d enqueued to queue.\n", k);
}

void dequeue() {
    if (isEmpty() == 1) {
        printf("Queue Underflow. Cannot dequeue.\n");
        return;
    }
    printf("%d dequeued from queue.\n", arr[front]);
    front = front + 1;
    count = count - 1;
}

int size() {
    return count;
}

void print() {
    int i;
    if (isEmpty() == 1) {
        printf("Queue is empty.\n");
        return;
    }
    printf("Queue (front to rear): ");
    for (i = front; i <= rear; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");
}

int main() {

```

```
create();

enqueue(10);
enqueue(20);
enqueue(30);
print();

printf("Size = %d\n", size());

dequeue();
print();

if (isEmpty() == 1)
    printf("Queue is empty.\n");
else
    printf("Queue is not empty.\n");

return 0;
}
```